GE1354 2025-26 Project Guidelines

1. Project Goal

The goal of this project is to design and develop an **electronic product** using the **Micro: bit** and external accessories to address a specific real-world need or problem. Students will apply their knowledge of:

- Micro: bit programming,
- Electronic components, and
- Python programming.

The aim is to create an **innovative and functional solution**.

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2. Group Size

Each group will consist of **4 students**.

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3. Scope of the Project

The project can be **creative or functional** and may fall (**<u>but not limited to</u>**) under one of the following categories:

A. Entertainment and Educational Tools

- **Digital Pets**: Create an interactive virtual pet with visual, physical, and sound feedback.
- **Digital Games**: Create an interactive game using Micro: bit sensors and actuators, such as reaction time challenges, maze navigation, or motion-based games.
- **Electronic Musical Instrument**: Build an instrument that produces sounds based on user input, such as motion, buttons, or light.
- **Magic Game Box**: Design a magic or a memory game box with 3 different color LEDs and 3 switches (separated hardware provided).

B. Monitoring and Control Systems

- **Obstacle Radar**: Develop a radar system that detects obstacles and provides visual and sound alerts.
- **Home Security System**: Create a system to detect motion or intrusion and send alerts wirelessly.
- **Multi-Point Weather Monitoring System**: Build a system to collect and display weather data (e.g., temperature or humidity) from multiple locations.

C. Smart Automation Devices

- **Remote Home Automation System**: Design a system to remotely control and monitor home appliances using wireless communication between Micro: bits.
- **DIY Temperature-Controlled Fan System**: Create a fan system that adjusts speed or activates based on temperature levels. Separate fan is required.
- **Smart Lighting System**: Automate and control lighting based on ambient light levels or motion detection.

These project ideas are examples to inspire your work. You are encouraged to come up with **your own creative solutions** that meet the hardware and software requirements.

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4. Minimum Hardware Requirements

- The project must involve:
  - **At least 2 Micro: bits**,
  - **2 or more sensors**, and
  - **2 or more actuators**.
- The two Micro: bits can communicate with each other:
  - **Wirelessly**, or
  - Via **serial USB connection** with a computer.
- Each group will be provided with the following sensors and actuators:
  - **Ultrasonic Module**
  - **RGB LED Module Circuit Board**
  - **OLED Display**
  - **Servo Motor**
- The Micro: bit has built-in features, which include:
  - Motion Sensors (accelerometer), Compass, Push Buttons, Speaker, Touch Sensor
  - Light Sensor, Sound Sensor, Temperature Sensor, and External Pressure Sensor

If your project requires **special sensors or actuators**, please consult with me before proceeding.

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5. Programming Language

The main programming language for the project is **Python**.

6. Working Schedule

Weeks	Task / Activities	Location	Check Point
6-8	Problem identification and initial design	Home	Project proposal with problem statement and outline of solution
9	Project Consultation	Lab / Email	Evaluation of your project ideas
9	System design (hardware and software)	Home	Circuit diagram, component list, and program flowchart
10-12	Software and hardware integration	Home	Functional prototype with at least one implemented feature
13	Live demonstration	Lab	Working demonstration of all project features
13	Final presentation and report preparation	Lab / Lecture Theatre	Project presentation and report submission

7. Project Deliverables (Per Group)

Each group is responsible for submitting the following deliverables:

Deliverable	Deadline
Project Proposal	Week 8
Prototype Demonstration	Week 13
Project Report	Week 13
Project Presentation	Week 12 and Week 13

8. Notes on Grading and Expectations

- **Innovative Features**: Creative additions beyond the minimum requirements (e.g., unique sensor/actuator use, additional functionality).
- **Software Algorithms**: Clean, efficient, and well-documented Python code that demonstrates logical flow and advanced problem-solving.
- **Live Demonstration**: Hands-on demonstration showing a functional prototype, with smooth operation and clear explanations. **Presentation and Documentation**: Includes a clear project proposal, well-structured report with diagrams, and a professional presentation.

9. Assessment Rubric

Grade	Description
A+	- Multiple innovative features exceeding project requirements.
	- Advanced software algorithms with excellent optimization and creativity.
	- Outstanding live demonstration showcasing all features seamlessly.
	- Exceptional presentation and report with detailed, well-organized content and visual aids.
	- Clear evidence of teamwork and mastery of project concepts.
A	- Two innovative features beyond basic requirements.
	- Well-structured and efficient software algorithms.
	- Very good live demonstration covering all required features.
	- High-quality presentation and thorough documentation with clarity and detail.
A-	- One innovative feature implemented.
	- Efficient and functional software algorithms.
	- Good live demonstration meeting all project requirements.
	- Thorough presentation and documentation with minor gaps.
B+	- All project requirements met (e.g., 2 Micro:bits, 2 sensors, 2 actuators, wireless communication).
	- Excellent live demonstration of required features.
	- Strong presentation and well-organized documentation.
B	- All project requirements met.
	- Good live demonstration, though minor issues may be present.
	- Clear presentation and documentation with no major omissions.
B-	- Most project requirements met (e.g., missing one sensor or actuator).
	- Successful live demonstration, but with noticeable gaps in functionality.
	- Adequate presentation and documentation, but lacking depth or detail.
C+	- Partial project requirements met (e.g., one Micro:bit used or incomplete sensor/actuator usage).
	- Incomplete live demonstration with some features non-functional.
	- Basic presentation and documentation with limited explanation of design and implementation.
C-	- Minimal project requirements met (e.g., basic sensor/actuator implementation).
	- Basic presentation and documentation with significant gaps.
D	- Core functionality lacking (e.g., missing communication between Micro:bits or actuator usage).
	- Poor presentation and documentation with minimal effort.
F	- No submission or presentation.